Research Idea: Trust, Fairness and Accessibility in AI-Written Code

Research Methods in Software Engineering, Prof. Shlomo Mark · Draft of 8 November 2026 · Group: _____

This is a research proposal. We plan the study; we do not run it in this course.

---

1. The problem

Developers now write a large part of their code with AI code assistants. This raises questions that are not only technical but human: how far people trust AI, whether they treat its work fairly, and who pays when its work is worse.

- **Trust.** If developers trust AI too much, they may check its code less, and bugs and security problems can reach users.
- **Fairness.** Reviewers may judge the same code differently only because they believe AI wrote it. This is a bias, either against AI or in its favour, and it affects how AI-written work is accepted.
- **Accessibility.** If AI-written web code has more accessibility problems, people with disabilities pay the price, often without anyone noticing.

What is not known is whether these effects exist and how strong they are. We study all three with simple, measurable questions.

The claims in this section are our motivation. They have not yet been checked against the literature; this is done in the review floor (floor 4).

2. Research questions and hypotheses

RQ1. Is developers' trust in AI related to how much of its code they check?
H1: Developers who trust AI more check less of its code. H0: developers who trust AI more do not check less of its code.
Developers: professional developers in Israel (paid, at least one year of experience) who use AI code assistants. Trust: a score on a trust questionnaire. Check: reading the code line by line or running tests on it before merging, as a percentage of the AI code they receive.

RQ2. Do reviewers report a different number of defects when code is labelled as AI-generated?
H2: Reviewers report more defects when the code is labelled as AI-generated. H0: the label does not increase the number of defects reported.
Reviewers: professional developers. All get the same code; half, chosen at random, see the label "AI-generated", the other half see no label. Reviewers are not told what is being tested. A reported defect is a review comment that points to a bug or a security problem, classified by two independent raters who do not know the group.

RQ3. Is AI-written web code more or less accessible than code written by developers?
H3: AI-written web code has more accessibility violations than developer-written code. H0: AI-written web code does not have more accessibility violations than developer-written code.
Web code: web forms built from one public specification (a form challenge on the Frontend Mentor website). AI-written forms are generated by AI code assistants from that specification. Developer-written forms are solutions to the same challenge published on GitHub before 21 June 2022, when GitHub Copilot became available to all developers. Accessibility: the number of WCAG 2.1 AA violations per form, found by an automated checker.

All three questions and hypotheses passed our question checker (floor 2) on 8 November 2026.

3. Strategy, reasoning and price

We use the eight research strategies of the ABC framework (Stol and Fitzgerald, 2018) and its three goals: **A** generalization over actors, **B** precision and control, **C** realism of context. No strategy maximizes all three, so each choice has a price.

	Strategy	Why it fits (reasoning)	Price we accept
RQ1 relationship	**Sample study** (neutral setting): an online questionnaire.	The question asks whether two variables go together, with nothing manipulated. We want an answer that holds for many developers (goal A). We have no access to a company, so we cannot observe real work.	Little control and no context. "Checking" is self-reported, not observed. It cannot show that trust *causes* less checking, only that they are related.
RQ2 causality	**Laboratory experiment** (contrived setting): an online review task with two random groups.	We manipulate one variable (the label) and compare groups that differ only in it. Only a controlled design can show that the label causes a difference (goal B).	Low realism: a short task with a code sample is not a real code review in a real team. The result holds for the code samples used.
RQ3 relationship	**Combination:** a **sample study** of existing developer solutions on GitHub, plus controlled generation of AI forms (a **laboratory experiment** with tools as actors).	Real developer code from many authors (goal A), and a controlled, repeatable AI side: same specification, same prompts, same checker (goal B). No human participants are needed.	The two groups come from different settings, so the comparison is observational, not an experiment. It is confounded by who publishes, by learners among the authors, and by how each form was made. Results hold for one challenge only.

Risks we fix in planning (accidental limitations)

- **RQ1 and RQ2** run in one online session, so we recruit only once. The review task comes first and the trust questionnaire after it, so the questions about AI do not reveal the aim of the task.
- **Sample size:** we expect 30–60 participants, unpaid, recruited through our networks, LinkedIn and Israeli developer groups. This is small for RQ2; we calculate the needed sample before we start and report the sample as a limitation.
- **RQ2:** two or three code samples, not one; random assignment recorded; a final question to check whether people guessed the aim; a debriefing at the end, since the aim is hidden.
- **RQ3:** we name the checker and its version and use the same settings for both groups; we name the AI tools and prompts and use several prompts per tool; one solution per author; a rule for solutions that do not render; a stricter check with only solutions from before 29 June 2021, when Copilot's limited preview began.
- **Ethics:** the university ethics committee approves the study before any data is collected. RQ2 hides its aim, so it needs approval for this and a debriefing. For RQ3 we do not store usernames or other identifying details.

4. Open points for the next floors

- **Review (floor 4):** find the literature behind section 1, and justify the direction of H2 (does a label make reviewers stricter or more lenient?) and of H3.
- **Design and measurement:** choose the trust questionnaire, the code samples for RQ2, the AI tools and the accessibility checker.

Prepared with the help of Claude as a research assistant. The questions, strategies and decisions were checked and chosen by the group.